



HK IMO Training 2018-2019
Problem Set 7
Date due: 10 November 2018



Problem 29. Given a rhombus $ABCD$, find the locus of the points P lying inside the rhombus and satisfying $\angle APD + \angle BPC = 180^\circ$.

Problem 30. Let the medians of $\triangle ABC$ meet at M . Let D and E be distinct points on the line BC such that $DC = CE = AB$, and let P and Q be points on the segments BD and BE , respectively, such that $2BP = PD$ and $2BQ = QE$. Determine, with reasons, $\angle PMQ$.

Problem 31. Show that $n^n + (n+1)^{n+1}$ is composite for infinitely many positive integers n .

Problem 32. Two persons play the following game with integers. The initial number is 2019^{2019} . The players move in turns. Each move consists of subtraction of an integer between 1 and 2018 inclusive, or division by 2019, rounding down to the closest integer when necessary. The player who first obtains a non-positive integer wins. Which player has a winning strategy?